CCCA Cell-State Analysis — Zhang et al. 2019

Hematologic · Follicular Lymphoma

10x · 5,645 malignant cells · 4 samples · 2 patients (FL1018, FL2001; FL1018T2 transformed-DLBCL)

GO / NO-GO GATES

Gate 1 — mixing & continuity

GO

Harmony θ-sweep (5–20) + per-neighbourhood iLISI: mixing plateaus at θ=8 (same-sample nbr frac 0.97→0.46; iLISI sample 1.9/4, patient 1.5/2). Single connected kNN component; 0 detached islands. Patient-private FL clones integrate into one continuous manifold; signature geography is cell-cycle-independent.

Gate 2 — module geography

GO

Non-branching case: Gast/pSom overlap on one axis (spatial r = −0.19, no two competing arms); Crest an inverse/terminal gradient (Crest vs Gast −0.52, vs pSom −0.60; joint pSom&Crest-high enrichment 0.23 ≪ 1). Single Gast/pSom → Crest developmental axis present → GO.

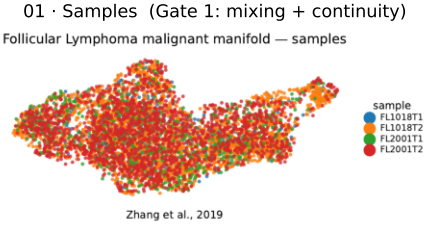
Gate 3 — cross-repression

NA

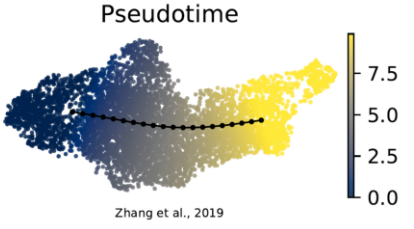
No bifurcation (curve, 0 forks) → branching-only gate not applicable. 04_correlation / 05_modules not produced.

STEP FIGURES

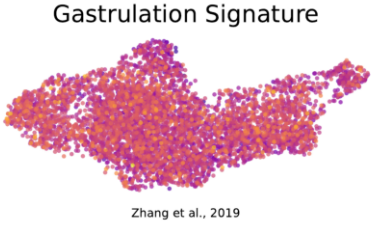
01 · Samples (Gate 1: mixing + continuity)



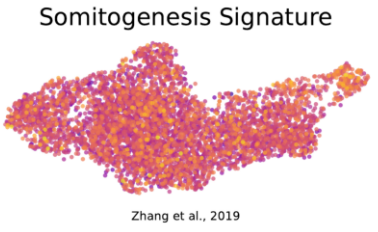
02 · Pseudotime manifold (linear, fold-free)



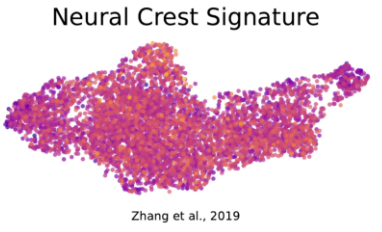
03 · Gastrulation signature



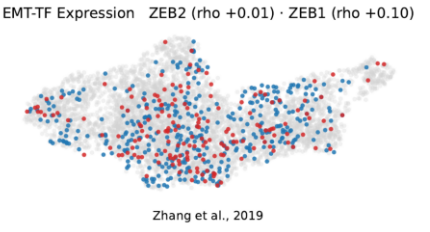
03 · Somitogenesis signature



03 · Neural-crest signature



04 · EMT-TF expression (ZEB2 · ZEB1)



Dataset: Follicular lymphoma malignant B cells; one continuous gastrulation→neural-crest axis, no arm split.

Verdict: **Pattern REPRODUCES as the non-branching case — GO / GO / NA (linear). No cross-repression hub.**

PARAMETER CHOICES & RATIONALE

Sample threshold	≥100 cells → all 4 kept (5,645 cells)	gap-finding: min sample 680 ≫ 2×floor, no under-populated sample
Overdispersed genes	636	scFates find_overdispersed (GAM residuals)
n_pcs	30	variance-ratio elbow past PC20; floored to robust 10x range
Integration (Harmony)	θ = 8 on sample	mixing plateau across θ=5–20 (SSF + iLISI); FL clones patient-private
Embedding	neighbors k=30, UMAP spread=1.3	+ radius-graph island removal (0 removed)
Geometry	Linear (non-branching)	Gast/pSom overlap, Crest inverse gradient
Topology	scFates curve, Nodes=20	0 forks, fold self-approach 2.96 (>1); PPT tree gave spurious forks
Root	Gast/pSom-high tip (Crest-low)	developmental order: Gast t≈0.04 → Crest 0.59 → pSom 0.82
Module means (raw)	Gast +0.012 · pSom +0.006 · Crest +0.009	honest per-cell score_genes readout
EMT-TFs shown	ZEB2 (27%, ρ +0.01) · ZEB1 (12%, ρ +0.10)	both terminus-associated; no root-end TF; VIM excluded (not a TF)